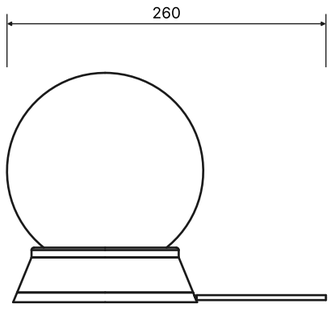
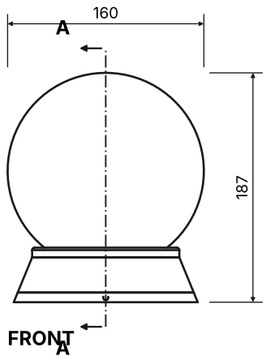
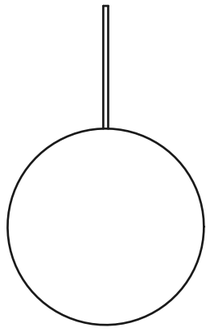
Scale 1:5. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

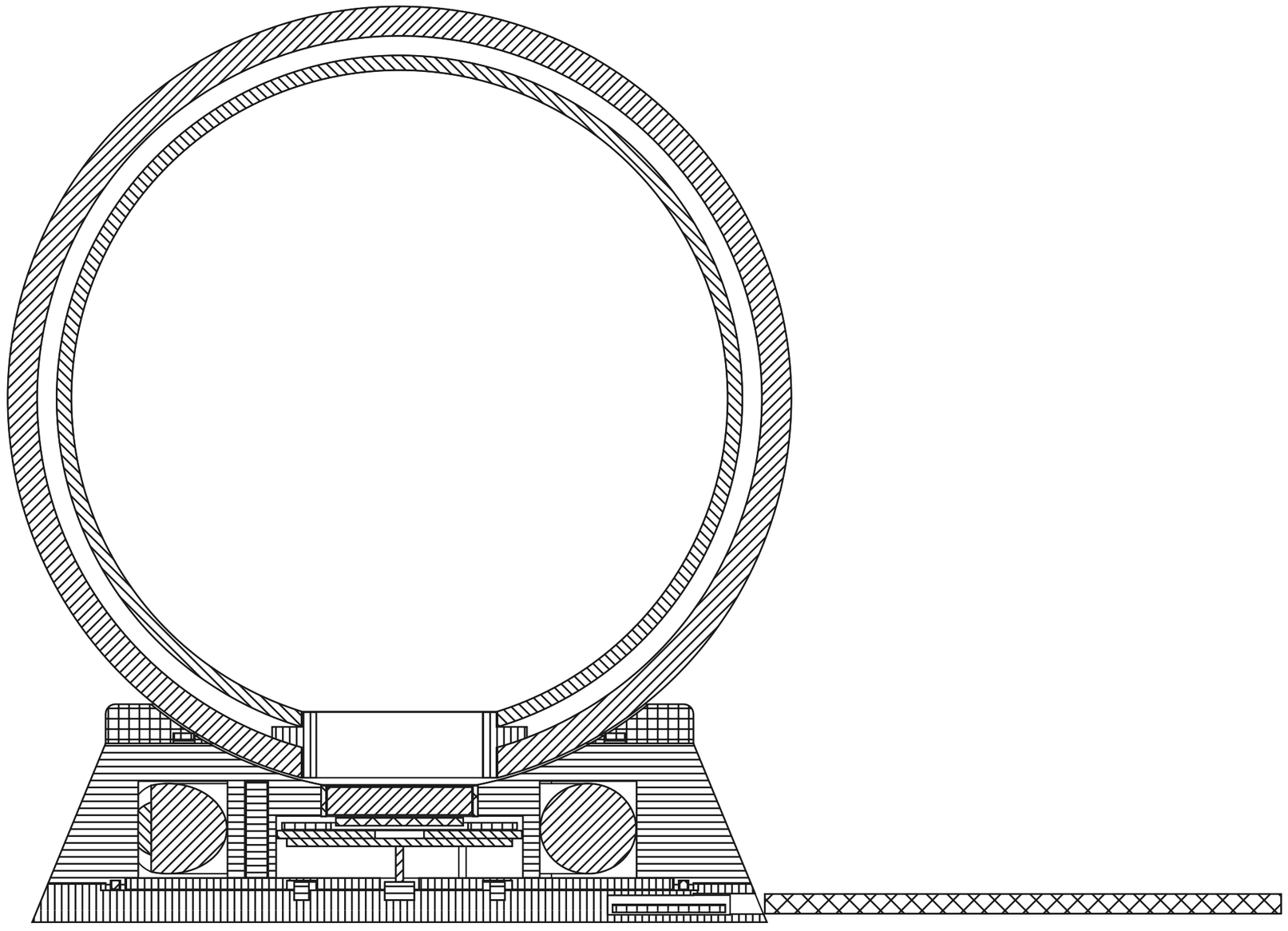
General arrangement

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. Section A-A (sheet 1): the product cut at x 0.0 and seen from the right; cut faces hatched, a different angle for each part.



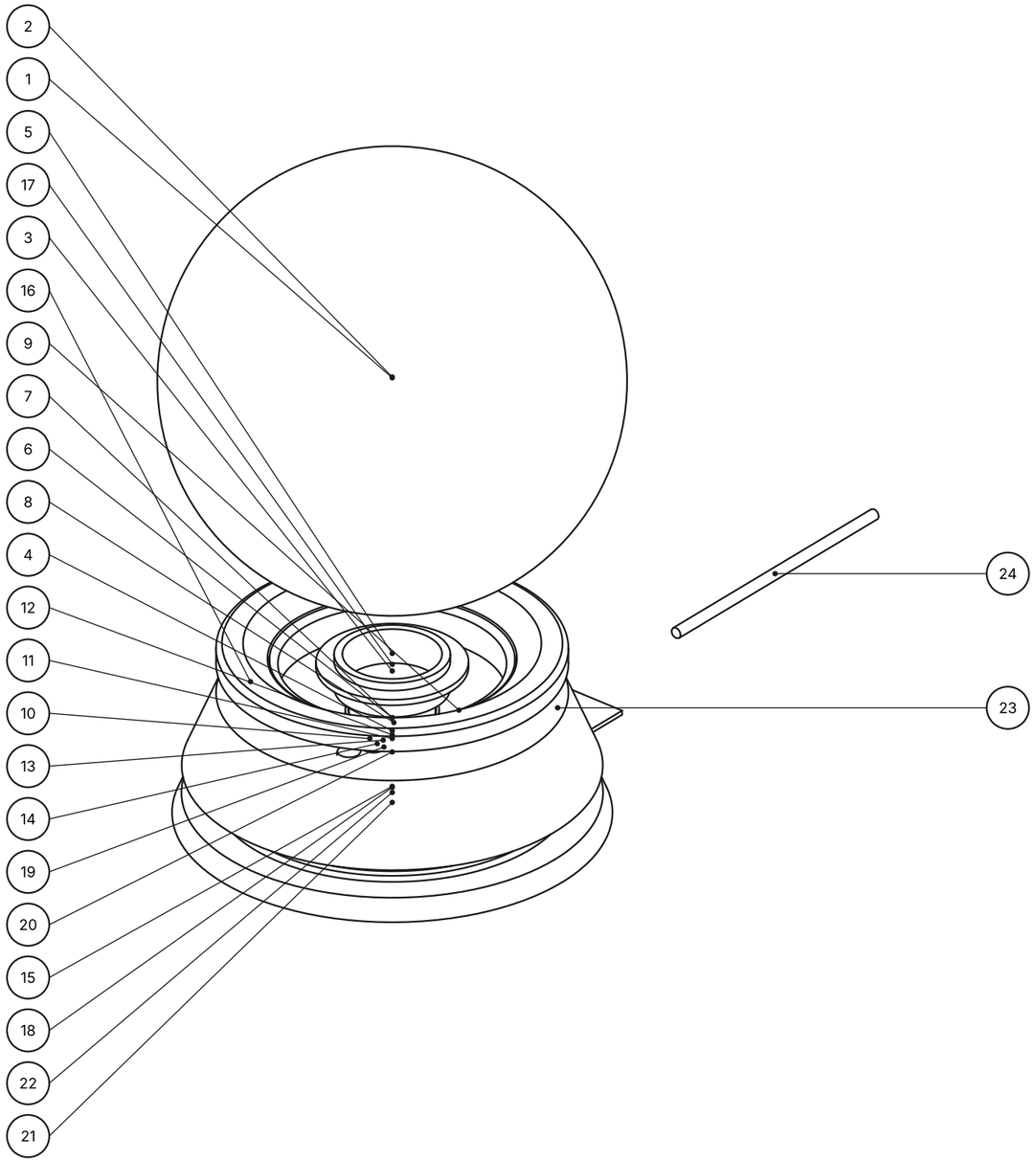
Section A-A

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

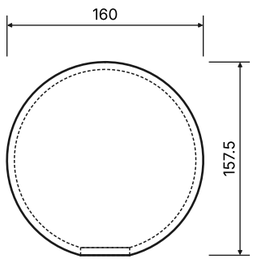
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

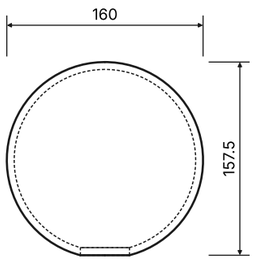
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

1. shell: formed, acrylic-trucast-formed; 440 cm3 each; 1 for the set; 1:5

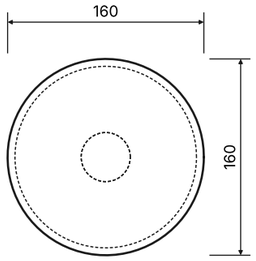
Core B (the owner, 2026-11-09): the outer ball, R 80 outside, R 74 inside, polished both faces (no frost). Two hemispheres drape-formed from 6 mm (1/4 in) cast sheet in the speaker's TruCast code for the tint (Ember 2423, Dust 2422, Acid 9097, Poison 2092, Ice 2069, Haze 2287, Fog 0000, Smoke 2074), the chart figure held at the 6 mm wall, on the existing female mould [check with the former: the 6 mm wall must form on the 9 mm wall's mould]; joined at the equator round the core with a polymerising acrylic cement (Acrifix 192 kind), the joint turned and polished; the Ø40 port cut at its foot



FRONT



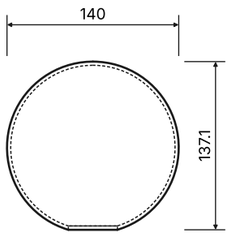
FROM THE RIGHT



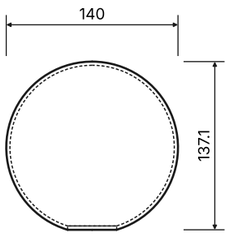
FROM ABOVE

2. core: formed, acrylic-opal-core; 173 cm3 each; 1 for the set; 1:5

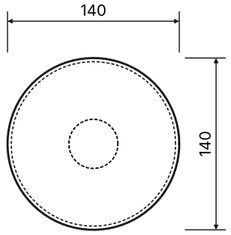
Core B: the core, a Ø140 ball of 3 mm white opal cast acrylic, Perspex GS Opal 041 (R 70 outside, R 67 inside), concentric with the outer ball (2026-11-10, the owner: "the same depth around the whole inner core"): a 4 mm gap all round, a 10 mm band of colour. Its own Ø40 opening at its foot over the window, its foot on the grommet's flange (its top at z 33.4), bonded on the grommet's 3 mm spigot (MMA). The former to confirm the 4 mm gap between the balls (±1 mm on both, the two equator seams in one plane); fallback Ø130, a 15 mm band. Captive once the outer halves are joined round it. …



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

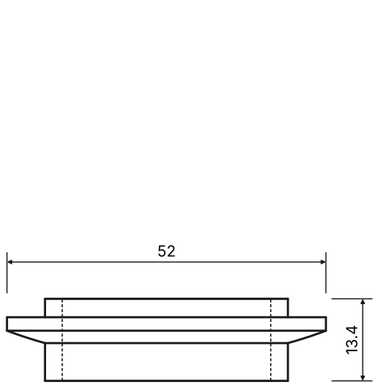
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

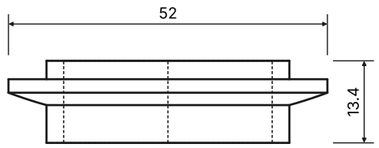
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

3. grommet: lathe, aluminium; 7 cm3 each; 1 for the set; 1:1

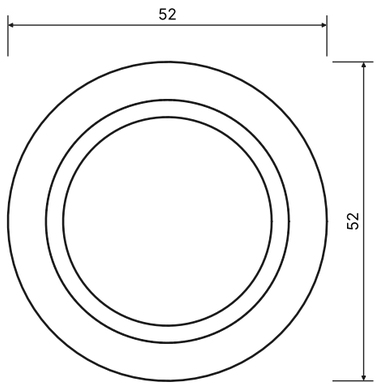
black anodised aluminium: lines the port, bonded with a two-part MMA adhesive (not cyanoacrylate). Review 2026-11-01, to draw with the dock: it grows a Ø38 × 6 spigot under the ball with three 30 deg L-slots, into a Ø38.4 socket round the window with three Ø3 stainless pins, so the ball stays captive in a drop and still swaps with a quarter turn (CORE-RECHARGE.md §6); Core B (2026-11-10, the concentric core): raised about 1.5 to suit. A flange Ø52 turned on it inside the ball, contoured underneath to R 74 to sit on the outer ball's inside (the axial stop), flat on top at z 33.4 under the ...



FRONT



FROM THE RIGHT



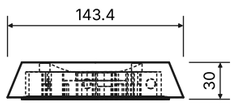
FROM ABOVE

4. base: lathe, cast-acrylic; 228 cm3 each; 1 for the set; 1:5

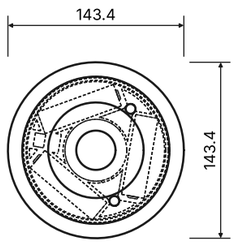
black cast acrylic, polished, turned from rod: 120 × 30 (round 32f's), the cup turned in it; from below the hatch's counterbore (112), the middle pocket (50) for the chassis, the board and the LED, the three cells' bays, the ballast's corners and the cable notch



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

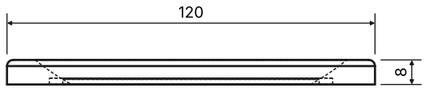
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

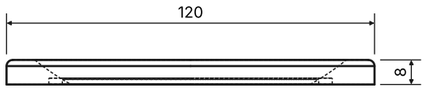
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

5. ring: lathe, aluminium; 36 cm3 each; 1 for the set; 1:2

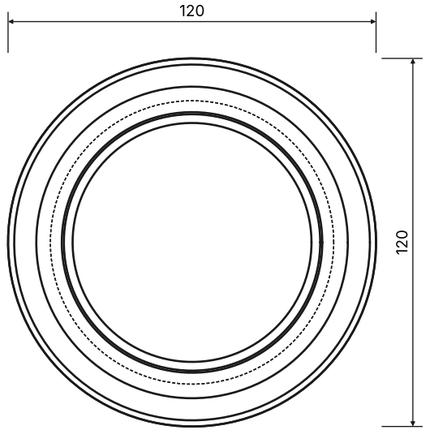
aluminium, black anodised, satin, insulated from the chassis: the touch (tap to switch, hold to dim, 2200 to 2700 K)



FRONT



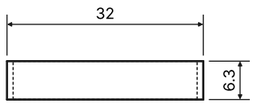
FROM THE RIGHT



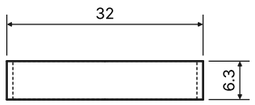
FROM ABOVE

6. well: lathe, ptfe; 1 cm3 each; 1 for the set; 1:1

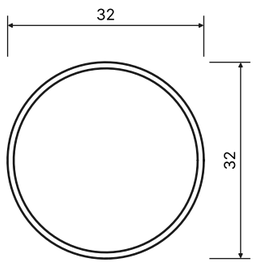
the light well (2026-11-10): a white PTFE sleeve, Ø32/Ø30 × 6, pressed into the base's bore round the window (the window's O-ring seals to it), so the walls round the LED reflect (about 0.95) where the black acrylic absorbed



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

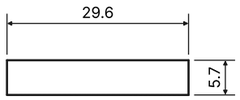
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

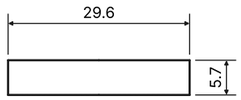
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

7. window: lathe, cast-acrylic; 4 cm3 each; 1 for the set; 1:1

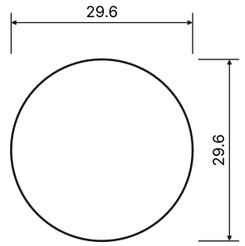
a clear acrylic plug, sealed with an O-ring: the LED under it, the base IP44 with the ball off



FRONT



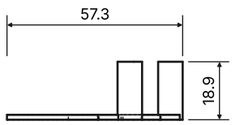
FROM THE RIGHT



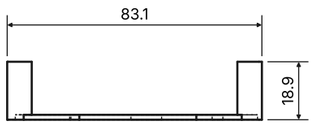
FROM ABOVE

8. chassis: mill, aluminium; 5 cm3 each; 1 for the set; 1:2

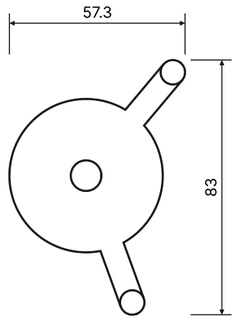
aluminium 6061, 50 across: the LED's heat spreader and the board's carrier, screwed up into the base on threaded inserts, an O-ring between. Review 2026-11-01: two arms 8 × 1.6 out to r 44 in the cell-free corners and a Ø8 post at each up to the touch ring (on a 0.5 insulating thermal pad): about 0.4 W/K to the ring, so about 4 K across the path at 1.5 W; the ring's 9,500 mm2 to the room about 16 K more [E]: the chassis about 20 K over the room at full while charging, to prove on the bench



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

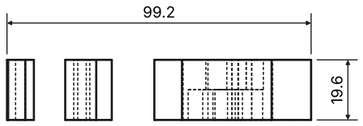
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

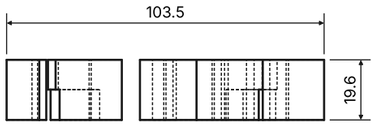
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

10. ballast: mill, steel; 17 cm3 each; 1 for the set; 1:2

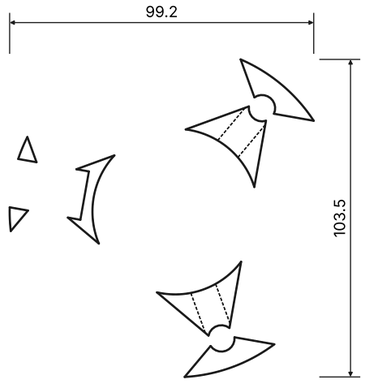
mild steel, zinc-plated: three corner pieces between the cells' bays, screwed up into the base



FRONT



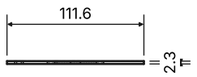
FROM THE RIGHT



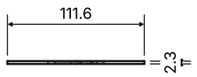
FROM ABOVE

15. hatch: mill, aluminium; 21 cm3 each; 1 for the set; 1:5

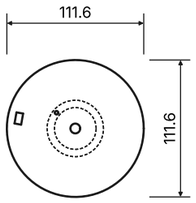
black anodised aluminium, 2.3 thick, in a counterbore under the base: six captive M3 screws (Phillips or Torx: commercially available tools; review 2026-11-01: six, was four, tapped into the steel ballast corners, not inserts in acrylic, nylon-patched), an O-ring (IP44); a label inside: the cells' type (LiFePO4 18650 only) and how to change them. Under it, to draw with the dock: a 0.5 304 stainless tray round the three bays (the bays 19.6 wide for it), so the cells sit in metal on every side (UL 153 SA; CORE-RECHARGE.md §6)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

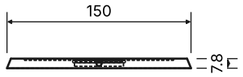
mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

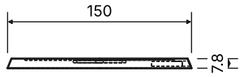
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

21. dock: lathe, cast-acrylic; 112 cm3 each; 1 for the set; 1:5

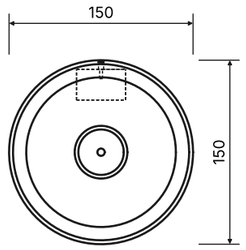
the contact dock (signed 2026-11-01): Ø150 × 8, polished black cast acrylic, turned; a 1.5 pocket Ø122 the base sits in; the contacts pressed in; a pocket at the back for its board and a channel for the cable; silicone dots underneath (CORE-RECHARGE.md)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	shell	formed	1	Cast acrylic sheet 6 mm, tinted through in the speaker's TruCast code, drape-formed over an aluminium mould, polished both faces	440 cm3, 170 × 170 × 167	1,534.01
2	core	formed	1	Cast acrylic sheet 3 mm, white opal (Hemisphere's diffuser's), formed over an aluminium core mould	173 cm3, 150 × 150 × 147	1,339.27
3	grommet	lathe	1	Aluminium 6061	62 × 62 × 23	47.25
4	base	lathe	1	Cast acrylic tube, tinted through	153 × 153 × 40	122.66
5	ring	lathe	1	Aluminium 6061	130 × 130 × 18	52.61
6	well	lathe	1	PTFE rod, white (virgin)	42 × 42 × 16	48.44
7	window	lathe	1	Cast acrylic tube, tinted through	40 × 40 × 16	45.98
8	chassis	mill	1	Aluminium 6061	67 × 93 × 29	49.52
9	pads	buy	1	Bergquist or 3M (to choose) Review 2026-11-01: two silicone gap pads Ø8 × 0.5, about 3 W/m K, electrically insulating (over 5 kV/mm), between the chassis's posts and the touch ring		1.00
10	ballast	mill	1	Mild steel	109 × 114 × 30	48.67
11	board	buy	1	to make (a contract PCB house) Core's board since round 41 (the 30 base), revised by the review of 2026-11-01: a USB PD sink at 9 V on the USB-C path and the dock's pins (an ideal diode on each input); the LTC4162-F LiFePO4 charger with power path		34.00
12	led	buy	1	Nichia or Seoul (to choose) A 30 mm round warm-dim LED board, two channels (2200 K and 2700 K, CRI 90+), 130 lm/W or better: 200 lm at 1.5 W or less (review 2026-11-01; was 2.5 W), on the aluminum chassis's heat path to the touch ring |		9.00
13	cells	buy	1	Power Sonic PSL-FP-IFR18650EC (round 42), three in parallel (1S3P): LiFePO4 18650 energy cell, 3.2 V, 1.5 Ah rated, 4.8 Wh each, 14.4 Wh the set; 2,000 cycles at 100 % depth of discharge; 65 × 18.2 mm (flat top on its datasheet: Core needs a		9.00
14	holder	buy	1	to make (in the milled base) No holder (review, 2026-11-01): bare spring contacts in the milled base. The − end has a nickel-plated conical spring in a pocket; the + end has a gold-flashed leaf at the bottom of an 8 mm shroud 1.5 deep, so only a		3.00
15	hatch	mill	1	Aluminium 6061	122 × 122 × 12	49.55
16	usb	buy	1	Two EPDM O-rings (the window, the chassis to the drum), a silicone foot ring 110 x 2 (Shore 40, non-slip), a sealed USB-C receptacle (IP67 rated, e.g. GCT USB4135 kind): IP44		5.00
17	seat	buy	1	A silicone seat ring 92 × 84 × 1.5 (Shore 40, black) in the base's cup: the ball rests on it, and it stops a nudge rolling it		1.50
18	foot-ring	buy	1	A silicone foot ring 117 × 114 × 2 (Shore 40, non-slip), and the hatch's O-ring		2.00
19	pins	buy	1	Mill-Max (to choose) Two gold-plated spring-loaded pins, about 1.5 mm travel, low force (about 0.8 N at 1.0 mm), on the pin board: + on the axis, - at r 20		3.00
20	pinboard	buy	1	to make (with the main board) A Ø46 × 1.6 board under the chassis carrying the dock's two pins, wired to the main board's dock input (its ideal diode and PD path)		4.00
21	dock	lathe	1	Cast acrylic tube, tinted through	160 × 160 × 18	103.23
22	dock-contacts	buy	1	to make (a turning shop) 304 stainless, turned and lapped: a post Ø6 (+) and a ring r 18.5 to 21.5 (-), pressed into the dock, 1.7 proud of its pocket		8.00

Parts list (1 of 2)

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

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Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
23	dock-board	buy	1	to make (a contract PCB house) The dock's board: a USB PD sink asking 9 V (STUSB4500 or CH224K class), a resettable fuse, a load switch that powers the contacts only when it sees the lamp's 47 kOhm sense resistor, a USB-C receptacle		6.00
24	dock-cable	buy	1	any (made to order) A slim black braided-fabric USB-C to USB-C cable, 2 m, 3 A, its plug into the dock's back edge (the dock signed 2026-11-01; the base's notch is closed); the same cable charges the lamp directly at its USB-C under the hatch		6.00
25	adapter	buy	1	any (UL/CE listed) A 20 W USB-C wall adapter, black, UL 62368-1 listed, DOE Level VI external power supply: it is part of Core's battery charger system as tested (the DOE standard and California Title 20, listed in the CEC's MAEDbS under Core's		7.00
26	screws	buy	1	Four captive M3 × 6 stainless pan-head screws (Phillips or Torx T10) with retaining washers, for the hatch		1.50
27	tray	buy	1	to make (a sheet-metal shop) Review 2026-11-01 (UL 153 SA): a 0.5 mm 304 stainless tray formed round Core's three cell bays and the chassis's corners, open only at the aluminium hatch, so the cells sit in metal on every side; it holds them if the		9.00
28	core-stock	buy	1	[option: stock core] a lighting-globe maker (to find) [check] OPTION (Core B, 2026-11-09): IN PLACE OF the formed core line, not on top of it. A stock Ø120 × 3 mm white opal acrylic sphere (the kind sold as a replacement lamp globe), if one		12.00

Parts list (2 of 2)

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-11-10 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. Core (the Orb until round 35): a crystal ball, a 160 tinted ball low in the cup of a 120 × 30 base (round 41), 180 tall. The ball lifts off: the LED is under a sealed window in the base, so balls swap colour by hand.
2. Core B, the owner's choice (2026-11-09): a ball in a ball. The outer ball is 6 mm tinted cast acrylic, polished both faces, two halves formed on the existing mould and joined at the equator; the Ø40 port at its foot. Inside it the core, a Ø140 ball of 3 mm white opal acrylic (Perspex GS 041, Hemisphere's opal), concentric with the outer ball (2026-11-10): a 4 mm gap, a 10 mm band of colour all round (fallback Ø130, a 15 mm band). Its own Ø40 opening sits on the grommet's flange and spigot: the LED lights the opal directly, and the core glows evenly. The core is captive once the outer halves are joined round it, and the grommet seals the gap for life. Design A (one 9 mm ball frosted inside) is in git and builds from orb_parts(O_A).
3. The same materials as the speakers: each tint's outer ball is cast in the speaker's TruCast code (Ember 2423, Dust 2422, Acid 9097, Poison 2092, Ice 2069, Haze 2287, Fog 0000, Smoke 2074), the chart figure held at the 6 mm wall: a dosed batch per code, cast to match where stock will not hold it; the chips decide. Haze 2287 may be translucent (Acme lists it "Violet - Translucent"): Haze is a custom cast for the speaker, and its outer ball would be too.
4. Cordless (the owner, 2026-10-31): three button-top LiFePO4 18650 cells (Power Sonic PSL-FP-IFR18650EC class, 14.4 Wh) on bare spring contacts in bays milled round the LED, behind a hatch underneath on four captive screws, the user's to replace with standard cells. It charges on its dock or from USB-C underneath at 9 V (USB PD, the 20 W adapter in the box; CORE-RECHARGE.md), and charges while it runs, about 2.7 hours to 90 % at 1.5 A. About five hours a charge (the copy until the bench proves more: computed, about 7.8 hours at full brightness, 1.5 W, at the default charge limit of 90 %, counted by a coulomb counter).
5. Long life: LiFePO4 keeps about 80 % of its capacity after about 2,000 cycles (ordinary Li-ion about 500); the charge limit, a battery-management circuit (protection, a coulomb counter, each cell behind back-to-back FETs off until checked, charging held off below 0 C and above 45 C by the cells' NTC, thermal foldback) and no exposed live contacts. To be listed as one product to UL 153 with Supplement SA; the cells to IEC 62133-2 and UN 38.3.
6. Touch the black ring: tap to switch, hold to dim, 2200 to 2700 K as it dims. IP44: the base sealed by O-rings at the window, the chassis and the hatch, the USB-C receptacle sealed; with the ball on or off.
7. Black, so the tint is the only colour: the base black cast acrylic, polished, over an aluminium chassis and a steel ballast; the ring, the port's lining and the hatch black anodised aluminium.

Notes

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